

MA511: Computer Programming

Practice Lab 2(Due: 09th Sep 2025)

To execute these labs, you will only use the tools/techniques we have discussed in class.

1. Write a recursive function to compute Fibonacci series. Write an iterative version and compare the results.
2. Write a program that reads an $n \times n$ matrix and:
 - a. Prints the main diagonal elements and their sum.
 - b. Prints the secondary diagonal elements and their sum.
 - c. Checks whether the matrix is diagonal or not
3. Write a recursive function to count the number of handshakes, if every person in a room shakes hand with every other person in the room. The function should print the number of times it is called.
4. Write a program to find the transpose of a square matrix.
5. Write a C program to test if a given square matrix is orthogonal (A square matrix S is said to be orthogonal if $SST = I$). The program should use a function with prototype
`int IsIdentityMatrix(int a[][SIZE]);`
The function will return 1, if the matrix $a[][]$ is an identity matrix and 0 if otherwise.

For example, if the input is $a[][2]=\{\{0, 1\},\{1, 0\}\}$ the output should be TRUE.

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